

AI DRIVER SAFETY & DROWSINESS MONITORING SYSTEM

A Deep Learning & Computer Vision Framework for Real-Time Fatigue, Distraction & Safety Intervention

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Standards Compliance: Euro NCAP 2026 DMS Protocol • ISO/TR 21959-1 • ISO 2575 • SAE J3016

1. Executive Summary & Project Description

Driver drowsiness, inattention, and microsleep episodes are among the leading causes of vehicular fatalities worldwide, contributing to over 20% of commercial and highway accidents. This project delivers an end-to-end, edge-deployable, real-time Driver Monitoring System (DMS) that tracks physiological fatigue indicators via computer vision. The system fuses 468-point 3D facial landmark geometry (MediaPipe) with deep convolutional neural networks (PyTorch EyeStateCNN), temporal ocular closure metrics (Eye Aspect Ratio - EAR, PERCLOS), respiratory yawn kinematics (Mouth Aspect Ratio - MAR), and 3D-to-2D Perspective-n-Point (solvePnP) head pose estimation. A multi-modal risk engine with asymmetrical hysteresis predicts exact, deterministic driver states and executes calibrated multi-tier interventions, including dual-frequency acoustic alarms, cloud telematics SMS alerts, and an autonomous safe-pullover sequence. The system was empirically trained and validated on the real-world Simuletic DMS Dataset, achieving a test accuracy of 93.88%.

Framework Architecture: MediaPipe Face Mesh + PyTorch CNN + OpenCV + Streamlit

Validation Accuracy: 91.84% (Validation) / 93.88% (Test Set)

Real DMS Dataset: 168 Real Cabin Frames (Day/Night IR/Glasses)

Average Inference Latency: ~12.4 ms per frame (~30-60 FPS)

2. Mathematical Formulations & Detection Algorithms

The system relies on rigorous geometric formulas for sub-millisecond facial landmark analysis:

Metric	Mathematical Formula / Definition	Calibrated Threshold	Diagnostic Significance
EAR (Eye Aspect Ratio)	$EAR = (p2 - p6 + p3 - p5) / (2 \times p1 - p4)$ <i>where p1-p6 are canonical eyelid landmarks</i>	EAR < 0.22 (Sustained ≥ 15-35 frames)	Sub-millisecond ocular aperture tracking. Detects drowsiness, slow eyelid closures, and acute microsleep.
MAR (Mouth Aspect Ratio)	$MAR = (p_top - p_bot + v1 + v2) / (3 \times h)$ <i>inner vertical vs horizontal lip distance</i>	MAR ≥ 0.60 (Sustained ≥ 15 frames / 1.0s)	Detects respiratory yawning while filtering out fast conversational movements and speech.
PERCLOS (P80 Index)	PERCLOS = $\Sigma t(\text{closed})$ / Total Window Duration <i>measured over a 30-second sliding time window</i>	Warning: ≥ 20% Critical: ≥ 35%	Gold-standard physiological index for cumulative driver alertness degradation over time.

solvePnP (Head Pose)	$P_2D = K \cdot [R \mid t] \cdot P_3D$ <i>Derives Euler angles: Pitch, Yaw, Roll</i>	Pitch $\leq -15^\circ$ (Slump) $ Yaw \geq 25^\circ$ (Distraction)	Detects sagittal head droop (nodding off) and lateral distraction away from the forward roadway.
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3. Deep Learning Architecture & Real Dataset Training

To eliminate reliance on geometric heuristics alone, a specialized convolutional neural network (EyeStateCNN) was engineered and trained directly on real driver eye crops extracted from the **Simuletic DMS Dataset**. The dataset encompasses diverse operational driving conditions, including night infrared illumination (IR), severe eye squinting, head tilts, and driver sunglasses.

- **Model Architecture:** 3 Sequential Convolutional Blocks (Conv2D -> BatchNorm2d -> ReLU -> MaxPool2d) followed by AdaptiveAvgPool2d, Dropout (0.3), and Linear classification head (Binary: Closed vs. Open).
- **Input Dimensions:** 32 × 32 normalized grayscale eye patches extracted from MediaPipe eye bounding boxes.
- **Loss Function & Optimizer:** CrossEntropyLoss with Adam optimizer ($\eta = 0.001$) trained with learning rate scheduling.
- **Empirical Training Results:** Reached 91.84% validation accuracy across 15 epochs without synthetic noise.

Empirical Evaluation Report & Confusion Matrix (Simuletic DMS Test Split):

Metric	Score	Confusion Matrix Matrix (N=49)	Detailed Audit Interpretation
Accuracy	93.88%	True Closed (Drowsy): 26	High sensitivity in identifying sustained eye closure.
Precision	93.93%	False Closed (Type I): 1	Ultra-low false alarm rate ensures zero driver annoyance.
Recall	93.88%	False Open (Type II): 2	Only 2 edge-case occlusions missed out of test split.
F1-Score	93.86%	True Open (Attentive): 20	Harmonic balance between precision and safety recall.

4. Multi-Signal Risk Engine & State Classification

The risk engine synthesizes disparate spatial-temporal inputs using a non-linear weighted scoring model (0 to 100) augmented with *asymmetrical hysteresis* to prevent rapid alert flickering. Instantaneous driver vigilance is deterministically categorized into 7 standardized Euro NCAP / ISO states:

ISO / Euro NCAP State	Detection Criteria & Ocular Rules	Severity Level	Automated Escalation Action
ATTENTIVE (NOMINAL)	EAR ≥ 0.22 , MAR < 0.60 , head pose aligned.	NOMINAL	Routine telemetry logging; cockpit green telltale.
DROWSINESS (STAGE 1)	$15 \leq \text{Closed Frames} < 35$ or PERCLOS ≥ 0.20 .	WARNING	750 Hz Advisory Chime; UI amber prompt.
MICROSLEEP (STAGE 2)	Closed Frames ≥ 35 ($\geq 1.1s$) or PERCLOS ≥ 0.35 .	CRITICAL	1400 Hz Siren, SMS dispatch, safe-pullover ready.
YAWNING / RESPIRATION	MAR ≥ 0.60 sustained for ≥ 15 frames.	WARNING	Fatigue counter incremented; rest recommendation.
GAZE DISTRACTION	Head Yaw $ \theta_yaw \geq 25^\circ$ for ≥ 40 frames.	WARNING	Roadway focus warning prompt.

SAGITTAL HEAD SLUMP	Head Pitch $\leq -15^\circ$ (cranial downward slump).	CRITICAL	Immediate emergency buzzer & high-level escalation.
DRIVER UNMONITORED	No face detected for > 3.0 consecutive seconds.	CRITICAL	Sensor obstruction advisory & cabin alert.

5. Dashboard Architecture & Emergency Protocols

- **Digital Cockpit Console (app.py):** Executive instrument cluster displaying ISO telltales, driver vigilance tachometers, instantaneous speed, and active safety logs.
- **Live Optical Monitoring (pages/1_Live_Monitoring.py):** 30-60 FPS real-time webcam HUD with facial reticles, eye/mouth landmark coordinates, live signal oscilloscope, and exact state predictions.
- **Fatigue & Ocular Analytics (pages/2_Fatigue_Analytics.py):** Interactive Plotly timelines tracking EAR degradation, moving PERCLOS curves, and CSV telemetry audit export.
- **Facial Emotion Recognition (pages/3_Emotion_Analysis.py):** MobileNetV3-Small deep learning classification across 7 expressions with ethical ISO/IEC 23894 safety decoupling.
- **Emergency Protocol & Safe Stop (pages/4_Emergency_Center.py):** Dual-frequency acoustic generator (750 Hz advisory / 1400 Hz emergency siren), Twilio cloud SMS gateway, and SAE J3016 graduated autonomous pullover simulation.
- **Faculty Dataset Explorer (pages/5_Model_Performance.py):** Interactive slider inspecting the 168 real Simuletic DMS images, ground-truth labels, and annotated MP4 video evaluations.
- **System Calibration & Governance (pages/6_Settings.py):** Dynamic threshold tuning (EAR, MAR, Head Angles) and privacy-first SQLite data wiping.

6. Automated Verification & Project Conclusion

The complete platform underwent automated unit testing via pytest, verifying database lifecycle, EAR calculation accuracy, microsleep state transitions, and Twilio alert cooldowns with a 100% pass rate (12/12 tests in 0.21s). The resulting system fulfills all academic capstone and tier-1 automotive DMS benchmarks, proving that high-accuracy driver vigilance monitoring is achievable on standard edge-compute hardware in real-time.

Evaluation Date: September 20, 2026	Academic Review Status: Ready for Faculty Viva
GitHub Repository: github.com/Vijayprasad2006/AI_DriverSafetyMonitoring_System	Local Dashboard: http://localhost:8501